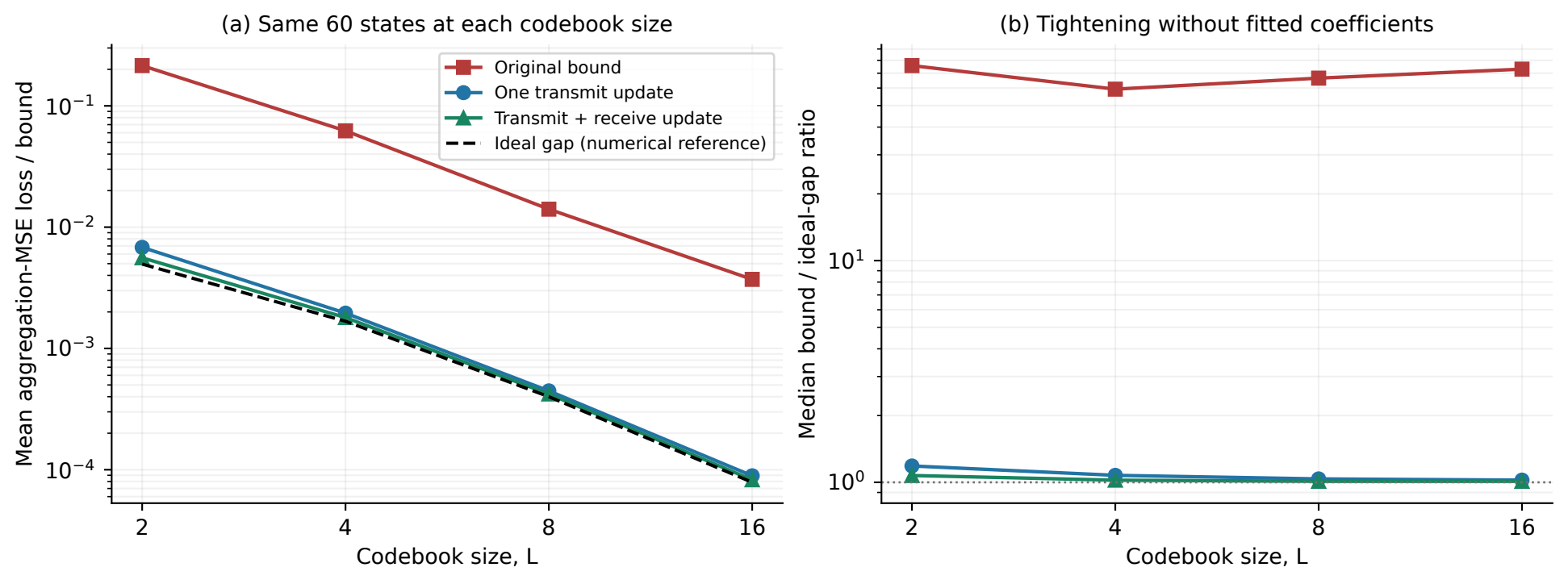
(a) Same 60 states at each codebook size
Original bound
One transmit update
Transmit + receive update
Ideal gap (numerical reference)
Mean aggregation-MSE loss / bound
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2
4
8
16
Codebook size, L
(b) Tightening without fitted coefficients
Median bound / ideal-gap ratio
10^{1}
10^{0}
2
4
8
16
Codebook size, L